Project Report: California House Price Predictor

AI & ML Task 1 Deliverable Report

Linear Regression **Scikit-Learn** **California Housing Dataset** **Model Pickle Verified**

1. Executive Summary & Objective

This project completes **Task 1: Build & Evaluate a Linear Regression Model (House Price Predictor)**. An end-to-end supervised machine learning pipeline was constructed using Python, Pandas, Scikit-Learn, Matplotlib, and Seaborn to predict median district housing values (MedHouseVal in $100,000s) based on demographic, structural, and geographic parameters.

2. Dataset Overview & Data Quality Audit

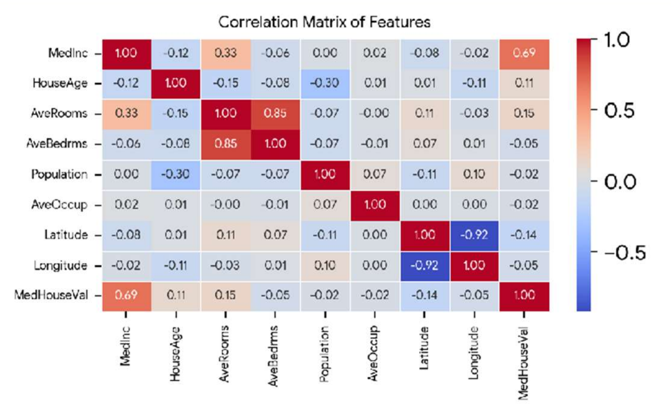
Dataset Dimensions & Schema

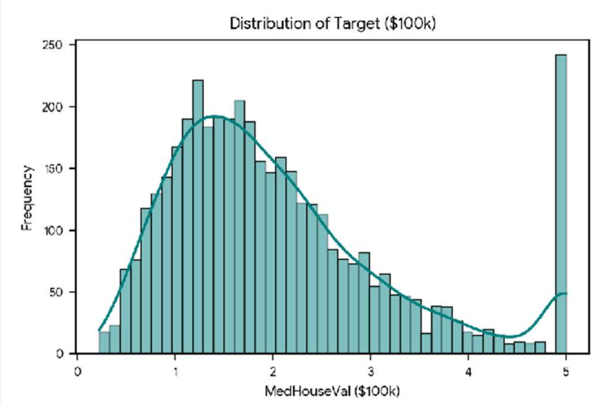
- **Total Records:** 20,640 rows × 9 columns (8 features + 1 target).
- **Features:** `MedInc`, `HouseAge`, `Ave Rooms`, `AveBedrms`, `Population`, `Aveo cup`, `attitude`, `Longitude`.
- **Target:** `MedHouseVal` (Median house value in $100,000s).

Preprocessing & Split Strategy

- **Missing / Null Values:** 0 nulls across all 9 features.
- **Train / Test Split:** 80% Training (16,512 samples) and 20% Testing (4,128 samples).
- **Random State:** Fixed at `42` for exact reproducibility.

3. Exploratory Data Analysis (EDA)





Key EDA Insights & Statistical Findings

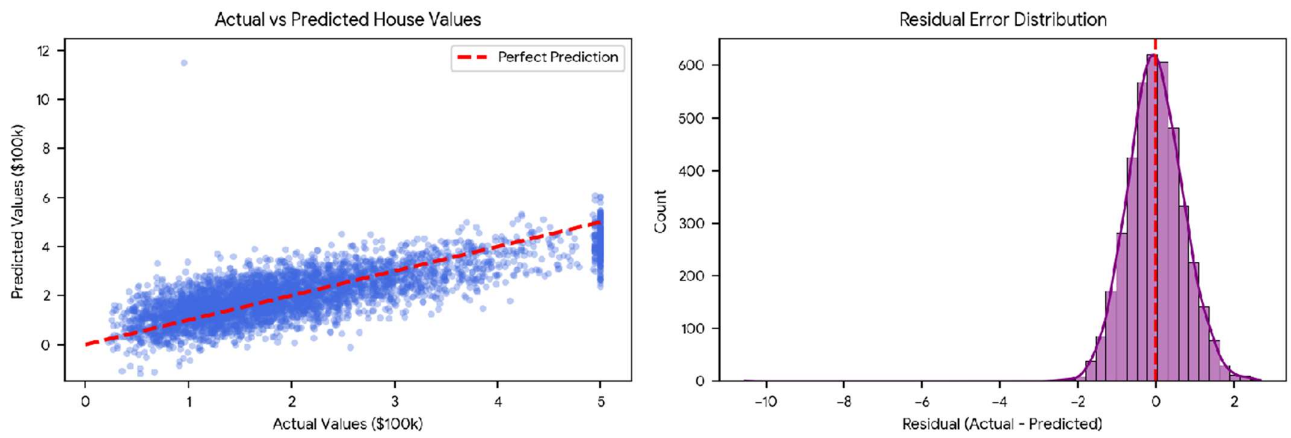
- **Primary Linear Driver:** Median Income (MedInc) exhibits the strongest positive linear correlation (**r = 0.69**) with housing price, serving as the dominant linear predictor.
- **Multi-collinearity:** `AveRooms` and `AveBedrms` have high correlation (**r = 0.85**), as larger residential units inherently possess more bedrooms.
- **Geospatial Dynamics:** `Latitude` and `Longitude` show a strong inverse correlation (**r = -0.92**) mapping California's geography, with non-linear coastal premiums.
- **Target Distribution Cap:** `MedHouseVal` is right-skewed with a prominent threshold spike at **$5.00001 ($500,000)** representing the census survey top-coding cap.

1. Model Performance & Evaluation Metrics

The Multiple Linear Regression model was fitted on 16,512 training instances and evaluated on 4,128 unseen test instances:

MEAN ABSOLUTE ERROR (MAE)	ROOT MEAN SQUARED ERROR (RMSE)	R^2 SCORE (VARIANCE EXPLAINED)
0.5332	**0.7456**	**57.58%**
≈ $53,320 avg error	≈ $74,560 penalty error	(R^2 = 0.5758)

2. Diagnostic Visualizations & Residual Analysis



Model Learned Coefficients

Feature	Coefficient	Direction
`AveBedrms`	**+0.783145**	+ Positive
`MedInc`	**+0.448675**	+ Positive
`HouseAge`	**+0.009724**	+ Positive
`Population`	**-0.000002**	- Negative
`AveOccup`	**-0.003526**	- Negative
`AveRooms`	**-0.123323**	- Negative
`Latitude`	**-0.419792**	- Negative
`Longitude`	**-0.433708**	- Negative

Diagnostic Observations

- **Residual Distribution:** Symmetrically centered at 0 with thin tails, confirming the baseline model provides unbiased central predictions.
- **Cap Saturation Band:** A prominent vertical cluster at actual value 5.0 reflects the $500k census clipping threshold.
- **Model Artifacts:** Occasional extreme positive outliers (e.g. predicted values > 10.0) occur when average rooms/ bedrooms ratios are anomalous in sparse rural blocks.

3. Model Persistence & Future Recommendations

- **Artifact Persistence:** The trained model was serialized and verified using `pickle.dump(model, f)` to `linear_regression_model.pkl` and reloaded successfully.
- **Non-Linear Spatial Modeling:** Incorporating tree-based ensembles (Random Forest, XGBoost, LightGBM) will effectively capture complex non-linear spatial interactions between latitude, longitude, and coastal proximity.
- **Feature Engineering:** Creating ratio features such as `rooms_per_household = AveRooms / AveOccup` and `bedrooms_ratio = AveBedrms / AveRooms` will resolve multi-collinearity.
- **Regularization:** Applying Ridge or ElasticNet regularization will penalize opposing collinear weights on room features.